이론시험 문제
해설

주의. 채점 기준의 배점은 나누어 부여하지 않는다!

문제 1 (10.0점)

문제 1A (3.0점)

Δt ΔN .

$\Delta p = \Delta N m u$,

$$F = \frac{\Delta p}{\Delta t} = \frac{\Delta N m u}{\Delta t} = n m u . \quad (1)$$

α F

$$A = Fl \sin\alpha . \quad (2)$$

l .

$$W = Mgl(1-\cos\alpha) . \quad (3)$$

$A = W$

$$\alpha_{\max} = 2\,\mathrm{arctg}\left(\frac{F}{Mg}\right) = 2\,\mathrm{arctg}\left(\frac{nmu}{Mg}\right) = 0.2 \text{ rad} = 11.65^\circ . \quad (4)$$

평가 항목	배점
(1) $F = \frac{\Delta p}{\Delta t} = \frac{\Delta N m u}{\Delta t} = n m u$	0,5
(2) $A = Fl \sin\alpha$	0,5
(3) $W = Mgl(1-\cos\alpha)$	0,5
(4) $\alpha_{\max} = 2\,\mathrm{arctg}\left(\frac{F}{Mg}\right) = 2\,\mathrm{arctg}\left(\frac{nmu}{Mg}\right)$	1,0
$\alpha_{\max} = 0.2 \text{ rad} = 11.65^\circ$	0,5
합계	**3,0**

문제 1B (4.0점)

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$$\frac{5}{2}P_1V_1 + \sigma 8\pi R_1^2 + \frac{kq^2}{2R_1} = \frac{5}{2}P_2V_2 + \sigma 8\pi R_2^2 + \frac{kq^2}{2R_2} \quad (1)$$

$$p_1 = \frac{4\sigma}{R_1} . \quad (2)$$

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